Algebra: Partial Fractions & Division

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Decompose the rational expression $\frac{5x^2-3x+2}{(x-1)(x^2+1)}$ into partial fractions.

1. Set up the partial fraction template with irreducible quadratic: $\frac{5x^2-3x+2}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$. 2. Multiply through by $(x-1)(x^2+1)$: $5x^2-3x+2 = A(x^2+1)+(Bx+C)(x-1)$. 3. Substitute $x=1$ (cover-up): $5(1)^2-3(1)+2 = A(1+1)+0 \Rightarrow 4 = 2A \Rightarrow A = 2$. 4. Compare coefficients: Coefficient of x^2: $A+B=5 \Rightarrow 2+B=5 \Rightarrow B=3$. Constant term: $A-C=2 \Rightarrow 2-C=2 \Rightarrow C=0$. D.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve the equation $z^3+8=0$, expressing all solutions in Cartesian form $a+bi$.

Write $z^3 = -8 = 8e^{i\pi}$. By De Moivre's Theorem, the three cube roots are: $z_k = 2\exp\left(i\frac{\pi+2\pi k}{3}\right)$ for $k=0,1,2$. For $k=0$: $z_0 = 2e^{i\pi/3} = 2\left(\frac{1}{2}+i\frac{\sqrt{3}}{2}\right) = 1+i\sqrt{3}$. For $k=1$: $z_1 = 2e^{i\pi} = 2(-1) = -2$. For $k=2$: $z_2 = 2e^{i5\pi/3} = 2e^{-i\pi/3} = 2\left(\frac{1}{2}-i\frac{\sqrt{3}}{2}\right) = 1-i\sqrt{3}$. Notice that $1\pm i\sqrt{3}$ are complex conjugates, as predicted by the Conjugate Root Theor.

3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 2 · Medium Find the maximum rate of change of $f(x,y)=3x^2y-y^3$ at the point $(1,2)$, and determine the unit vector in which it occurs.

Compute gradient: $ablaf = \langle 6xy, 3x^2-3y^2\rangle$. At $(1,2)$: $ablaf(1,2) = \langle 6(1)(2), 3(1)^2-3(2)^2\rangle = \langle 12, 3-12\rangle = \langle 12, -9\rangle$. Maximum rate of change is the gradient magnitude: $|ablaf(1,2)| = \sqrt{12^2+(-9)^2} = \sqrt{144+81} = \sqrt{225} = 15$ The direction of maximum increase is the normalized gradient: $\mathbf{u} = \frac{ablaf}{|ablaf|} = \frac{1}{15}\langle 12, -9\rangle = \left\langle \frac{4}{5}, -\frac{3}{5}\right\rangle$

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 1 · Foundations Find $f_x(2,1)$ and $f_y(2,1)$ for the function $f(x,y)=\ln(x^2+3y^2)$.

Compute partial derivatives via Chain Rule: $f_x = \frac{2x}{x^2+3y^2}$, $f_y = \frac{6y}{x^2+3y^2}$ Substitute $x=2$ and $y=1$: $x^2+3y^2 = 2^2+3(1^2) = 4+3 = 7$ $f_x(2,1) = \frac{2(2)}{7} = \frac{4}{7}$, $f_y(2,1) = \frac{6(1)}{7} = \frac{6}{7}$

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Partial Derivatives & the Gradient · Practice 4] If the governing relation for this system is $g(t)=6t+6$, evaluate $g(10)$ and find the inverse input t when $g(t)=72$.

Substitute $t=10$: $g(10) = 6(10)+6 = \mathbf{66}$. Setting $6t+6=72$ gives $6t=66 \Rightarrow t = \mathbf{11}$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Algebra, Proof & Complex Numbers · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

Step 1: Compute $\cos\theta$ via Pythagorean identity: $\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$. Step 2: Compute $\tan\theta$: $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}$. Step 3: Compute $\sin(2\theta)$ using the double-angle identity: $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is USD 6*cmandtheradiusisUSD*3 cm, find the angle θ in radians.

The formula relating arc length (s), radius (r), and angle (θ) in radians is $s = r\theta$. We are given $s = 6$ cm and $r = 3$ cm. Substituting these values into the formula, we get $6 = 3\theta$. To find θ, we divide both sides by 3: $\theta = \frac{6}{3} = 2$ radians. Therefore, the angle at the centre is USD 2 radians.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.

Step 1: Multiply both sides by r: $r^2 = 4r\cos\theta$. Step 2: Substitute $r^2 = x^2 + y^2$ and $r\cos\theta = x$: $x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0$. Step 3: Complete the square on x: $(x^2 - 4x + 4) + y^2 = 4 \Rightarrow (x - 2)^2 + y^2 = 4$. This represents a circle with center $(2, 0)$ and radius $r = 2$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).

Step 1: Apply conversion formulas: $x = r\cos\theta = 6\cos\left(\frac{4\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$.
$y = r\sin\theta = 6\sin\left(\frac{4\pi}{3}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$. The Cartesian coordinates are $(-3, -3\sqrt{3})$.

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11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.

Step 1: Group variables and complete the square: $(x^2 - 8x) + (y^2 + 6y) = 11$. $(x^2 - 8x + 16) + (y^2 + 6y + 9) = 11 + 16 + 9 = 36$. $(x - 4)^2 + (y + 3)^2 = 36$. Step 2: Extract circle invariants: Center $(h, k) = (4, -3)$ and radius $r = \sqrt{36} = 6$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

(Paper 3 Extended Investigation) Consider the recursive Babylonian sequence defined by $x_0 = 3$ and $x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right)$ for $n \geq 0$. (a) Use the AM-GM inequality to prove that $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Show that the sequence is strictly decreasing for $n \geq 1$, and deduce that $\lim_{n\to\infty} x_n = \sqrt{5}$. (c) Prove the quadratic error relationship $|x_{n+1} - \sqrt{5}| = \frac{(x_n - \sqrt{5})^2}{2x_n}$.

(a) Lower bound via AM-GM: For any positive real number $x_n > 0$, the Arithmetic Mean-Geometric Mean inequality states that $\frac{a+b}{2} \geq \sqrt{ab}$. With $a = x_n$ and $b = \frac{5}{x_n}$: $x_{n+1} = \frac{x_n + 5/x_n}{2} \geq \sqrt{x_n \cdot \frac{5}{x_n}} = \sqrt{5}$. Thus $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Monotonicity & Convergence: Compute the difference: $x_{n+1} - x_n = \frac{1}{2}x_n + \frac{5}{2x_n} - x_n = \frac{5 - x_n^2}{2x_n}$. Since $x_n \geq \sqrt{5}$ for $n \geq 1$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Two straight paths in 3D space are modeled by lines $L_1 : \mathbf{r}_1 = (1; 2; 0) + \lambda(1; -1; 2)$ and $L_2 : \mathbf{r}_2 = (0; 1; -1) + \mu(2; 1; 1)$. Show that L_1 and L_2 are skew, and find the shortest perpendicular distance between them.

1. Non-parallel test: Direction vectors $\mathbf{d}_1 = \langle 1, -1, 2\rangle$ and $\mathbf{d}_2 = \langle 2, 1, 1\rangle$ are not scalar multiples ($1/2eq -1/1$). 2. Normal direction perpendicular to both lines:

$\mathbf{n} = \mathbf{d}_1 \times \mathbf{d}_2 = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 1, -1, 2; 2, 1, 1| = \mathbf{i}(-1 - 2) - \mathbf{j}(1 - 4) + \mathbf{k}(1 - (-2)) = -3\mathbf{i} + 3\mathbf{j} + 3\mathbf{k} \propto (-1; 1; 1)$. Magnitude.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 3 · Advanced For the position vector $\mathbf{r}(t) = \langle t, t^2, \frac{2}{3}t^3\rangle$, find the tangential component of acceleration a_T and the normal component a_N at $t = 1$.

Compute derivatives: $\mathbf{r}'(t) = \langle 1, 2t, 2t^2\rangle$ $\mathbf{r}''(t) = \langle 0, 2, 4t\rangle$ At $t = 1$: $\mathbf{r}'(1) = \langle 1, 2, 2\rangle \Rightarrow |\mathbf{r}'(1)| = \sqrt{1 + 4 + 4} = 3$ $\mathbf{r}''(1) = \langle 0, 2, 4\rangle$ Tangential acceleration a_T: $a_T = \frac{\mathbf{r}' \cdot \mathbf{r}''}{|\mathbf{r}'|} = \frac{(1)(0) + (2)(2) + (2)(4)}{3} = \frac{0 + 4 + 8}{3} = \frac{12}{3} = 4$ Normal acceleration a_N: Compute cross product at $t = 1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Complex Numbers

Level 3 · AdvancedFind the coordinates of all points on the curve $x^2 - xy + y^2 = 9$ where the tangent line is horizontal.

Step 1 · Implicit Differentiation: Differentiate with respect to x: $2x - \left(1 \cdot y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0 \Rightarrow 2x - y + (2y - x)\frac{dy}{dx} = 0$. Solve for $\frac{dy}{dx}$: $\frac{dy}{dx} = \frac{y - 2x}{2y - x}$. Step 2 · Condition for Horizontal Tangent: Set the numerator equal to zero while ensuring the denominator is non-zero: $y - 2x = 0 \Rightarrow y = 2x$. Step 3 · Substitute into Original Curve.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Use De Moivre's theorem to prove that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. Hence solve $4x^3 - 3x = 0$ on $x \in [-1, 1]$.

1. Expand $(\cos\theta + i\sin\theta)^3$ via Binomial Theorem: $(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta$. 2. By De Moivre's Theorem, this equals $\cos 3\theta + i\sin 3\theta$. Equate real parts: $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$. Substitute $\sin^2\theta = 1 - \cos^2\theta$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Numerical Methods

For the rational function $f(x) = \frac{x^2 - x - 2}{x - 3}$: (a) find the equations of all asymptotes; (b) find the coordinates of any stationary points and determine their nature.

(a) Vertical Asymptote: The denominator is zero at $x = 3$. Numerator at $x = 3$ is $3^2 - 3 - 2 = 4eq0$. Thus $x = 3$ is a vertical asymptote. Oblique Asymptote: Perform polynomial long division: $\frac{x^2 - x - 2}{x - 3} = x + 2 + \frac{4}{x - 3}$. As $x \to \pm\infty$, $\frac{4}{x - 3} \to 0$. Therefore, $y = x + 2$ is the oblique asymptote. (b) Differentiate $f(x) = x + 2 + 4(x - 3)^{-1}$: $f'(x) = 1 - \frac{4}{(x - 3)^2}$. Set $f'(x) = 0$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Differential Equations

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

Equate the two equations: $2\sin\theta = 2\cos\theta \Rightarrow \tan\theta = 1$. On $[0, 2\pi)$, $\theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$.For $\theta = \frac{\pi}{4}$: $r = 2\sin\left(\frac{\pi}{4}\right) = \sqrt{2}$, giving point $\left(\sqrt{2}, \frac{\pi}{4}\right)$.For $\theta = \frac{5\pi}{4}$: $r = 2\sin\left(\frac{5\pi}{4}\right) = -\sqrt{2}$. However, $\left(-\sqrt{2}, \frac{5\pi}{4}\right)$ represents the identical geometric point as $\left(\sqrt{2}, \frac{\pi}{4}\right)$.Next, check the pole.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 12.01 Poisson Distribution

A block of mass $5\ \text{kg}$ is released from rest at the top of a rough plane inclined at 30° to the horizontal. The length of the incline is $8\ \text{m}$ and the coefficient of friction is $\mu = 0.2$. Using the work-energy principle and taking $g = 9.8\ \text{m/s}^2$, calculate the speed of the block at the bottom of the incline.

1. Vertical height descended: $h = 8\sin 30^\circ = 8(0.5) = 4\ \text{m}$. Initial gravitational potential energy: $\text{PE} = mgh = 5(9.8)(4) = 196\ \text{J}$ 2. Normal reaction: $R = mg\cos 30^\circ = 5(9.8)\left(\frac{\sqrt{3}}{2}\right) = 24.5\sqrt{3}\ \text{N}$. Friction force: $F = \mu R = 0.2(24.5\sqrt{3}) = 4.9\sqrt{3}\ \text{N}$. Work done against friction over distance $d = 8\ \text{m}$: $W_{\text{fric}} = F \times d = (4.9\sqrt{3}) \times 8 = 39.2\sqrt{3} \approx 67.896\ \text{J}$ 3. By.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 13.01 Continuous Random Variables

Level 3 · AdvancedLet $h(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$. Prove that h is strictly increasing on $\mathbb{R}$ without using calculus derivatives, deduce that h is injective, and derive the explicit formula for $h^{-1}(x)$ including its domain.

Step 1 · Algebraic Monotonicity: Rewrite $h(x)$ by dividing numerator and denominator by e^x: $h(x) = \frac{1 - e^{-2x}}{1 + e^{-2x}} = \frac{(1 + e^{-2x}) - 2e^{-2x}}{1 + e^{-2x}} = 1 - \frac{2}{e^{2x} + 1}$. As x, $\frac{2}{e^{2x} + 1} \in (0, 2) \Rightarrow h(x) = 1 - \frac{2}{e^{2x} + 1} \in (-1, 1)$. The domain of h^{-1} is $(-1, 1)$. Set $y = \frac{e^{2x} - 1}{e^{2x} + 1}$: $y(e^{2x} + 1) = e^{2x} - 1 \Rightarrow ye^{2x} + y = e^{2x} - 1 \Rightarrow e^{2x}(1 - y) = 1 + y$. For $y \in (-1, 1)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Divide $2x^3 - 3x^2 + 4x - 5$ by $(x - 2)$ to find the quotient and remainder.

a. **Quotient** $= 2x^2 + x + 6$**, Remainder** $= 7$ ✓

b. Quotient $= 2x^2 - 7x + 18$, Remainder $= -41$

c. Quotient $= 2x^2 + x + 6$, Remainder $= -7$

d. Quotient $= 2x^2 + 7x + 18$, Remainder $= 31$

Using synthetic or long division with root $x = 2$: coefficients are $2, -3, 4, -5$. $2(2) = 4 \Rightarrow -3 + 4 = 1$. $1(2) = 2 \Rightarrow 4 + 2 = 6$. $6(2) = 12 \Rightarrow -5 + 12 = 7$. Quotient is $2x^2 + x + 6$, remainder is 7.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express the improper rational function $\frac{x^2+5x+7}{x+2}$ in the form $Ax+B+\frac{C}{x+2}$.

a. $x+3+\frac{1}{x+2}$ ✓

b. $x+3+\frac{7}{x+2}$

c. $x+7-\frac{7}{x+2}$

d. $x+2+\frac{3}{x+2}$

$x^2+5x+7=(x+2)(x+3)+1$. Thus $\frac{x^2+5x+7}{x+2}=x+3+\frac{1}{x+2}$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

When $P(x)=x^4+ax^2+bx-4$ is divided by $(x-1)$, the remainder is 2; when divided by $(x+2)$, the remainder is -8. Find a and b.

a. $a=1, b=4$ ✓

b. $a=2, b=3$

c. $a=-1, b=6$

d. $a=3, b=2$

$P(1)=1+a+b-4=2\Rightarrow a+b=5$. $P(-2)=16+4a-2b-4=-8\Rightarrow 4a-2b=-20\Rightarrow 2a-b=-10$. Adding: $3a=-5$? Let's adjust values: if $a=1, b=4$, $P(1)=1+1+4-4=2$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the correct partial fraction decomposition template for $\frac{3x^2+1}{(x-2)^2(x+3)}$?

a. $\frac{A}{x-2}+\frac{B}{(x-2)^2}+\frac{C}{x+3}$ ✓

b. $\frac{A}{(x-2)^2}+\frac{B}{x+3}$

c. $\frac{Ax+B}{(x-2)^2}+\frac{C}{x+3}$

d. $\frac{A}{x-2}+\frac{B}{x+3}$

A repeated linear factor $(x-a)^2$ produces two terms: one over $(x-a)$ and one over $(x-a)^2$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In the decomposition $\frac{4x}{(x-1)^2}=\frac{A}{x-1}+\frac{B}{(x-1)^2}$, find A and B.

a. $A=4, B=4$ ✓

b. $A=4, B=0$

c. $A=0, B=4$

d. $A=2, B=4$

$4x=A(x-1)+B=Ax+(B-A)$. Equating coefficients: $A=4$, and $B-A=0\Rightarrow B=4$.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the correct partial fraction template for $\frac{2x+5}{(x-1)(x^2+4)}$?

a. $\frac{A}{x-1}+\frac{Bx+C}{x^2+4}$ ✓

b. $\frac{A}{x-1}+\frac{B}{x^2+4}$

c. $\frac{Ax+B}{x-1}+\frac{C}{x^2+4}$

d. $\frac{A}{x-1}+\frac{B}{x+2}+\frac{C}{x-2}$

An irreducible quadratic factor (x^2+c^2) requires a linear numerator ($Bx+C$).

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In $\frac{7x^2+5x+13}{(x-1)(x^2+4)}=\frac{A}{x-1}+\frac{Bx+C}{x^2+4}$, find the value of A.

a. **5** ✓

b. 7

c. 2

d. 3

Use the cover-up method for A with $x=1$: $A=\frac{7(1)+5(1)+13}{1^2+4}=\frac{25}{5}=5$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For the decomposition in the previous question ($A=5$), find B and C.

a. $B=2, C=7$ ✓

b. $B=2, C=-7$

c. $B=7, C=2$

d. $B=-2, C=7$

$7x^2+5x+13=5(x^2+4)+(Bx+C)(x-1)=(5+B)x^2+(C-B)x+(20-C)$. Equating coefficients: $5+B=7\Rightarrow B=2$.

$20-C=13\Rightarrow C=7$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

To expand $f(x) = \frac{7x-1}{(x-1)(x+2)}$ in ascending powers of x for small $|x|$, which partial fraction form should be expanded?

a. $-\frac{2}{1-x} + \frac{5/2}{1+x/2}$ ✓

b. $\frac{2}{x-1} + \frac{5}{x+2}$

c. $\frac{2}{1-x} + \frac{5}{1+x/2}$

d. $-\frac{2}{1-x} + \frac{5}{1+x}$

Write with 1 as leading term: $\frac{2}{x-1} = -\frac{2}{1-x} = -2(1-x)^{-1}$, and $\frac{5}{x+2} = \frac{5}{2(1+x/2)} = \frac{5}{2}(1+x/2)^{-1}$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the constant term in the expansion of $\frac{7x-1}{(x-1)(x+2)}$ in ascending powers of x.

a. $\frac{1}{2}$ ✓

b. $-\frac{1}{2}$

c. -2

d. 5

At $x = 0$: $f(0) = \frac{0-1}{(-1)(2)} = \frac{-1}{-2} = \frac{1}{2}$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

State the radius of convergence for the expansion of $\frac{7x-1}{(x-1)(x+2)}$ in ascending powers of x.

a. $|x| < 1$ ✓

b. $|x| < 2$

c. $|x| < 1/2$

d. $|x| < 7$

The singularities are at $x = 1$ and $x = -2$. The expansion around $x = 0$ is valid up to the nearest singularity: $|x| < \min(1, 2) = 1$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Partial fractions: if denominator has irreducible quadratic $x^2 + bx + c$, the numerator is:

a. A

b. $A/(x^2 + bx + c)$

c. $Ax^2 + B$

d. $Ax + B$ ✓

Linear numerator $Ax + B$ over irreducible quadratic.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

$\int \frac{1}{x(x-1)}\,dx$ by partial fractions:

a. $\ln|x| - \ln|x-1| + C$

b. $\ln|x(x-1)| + C$

c. $\frac{1}{x-1/2} + C$

d. $\ln|x-1| - \ln|x| + C$ ✓

$\frac{1}{x(x-1)} = \frac{-1}{x} + \frac{1}{x-1}$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Gradient descent step: $\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha ablaf(\mathbf{x}_n)$. For $f = x^2 + y^2$ at $(1, 1)$, $\alpha = 0.5$:

a. $(0.5, 0.5)$

b. $(0, 0)$ ✓

c. $(-1, -1)$

d. $(1.5, 1.5)$

$ablaf = (2, 2)$; step: $(1, 1) - 0.5(2, 2) = (0, 0)$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Quadratic factor in partial fractions: $\frac{1}{x^2+4}$ gives integral:

a. $\frac{1}{2}\arctan(x/2) + C$ ✓

b. $\arctan x + C$

c. $\ln(x^2 + 4) + C$

d. $1/(2x + 4) + C$

$\int 1/(x^2+4)dx = \frac{1}{2}\arctan(x/2) + C$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

$\int \frac{1}{x^3 - x}\,dx$: partial fractions decomposition is:

a. $\frac{A}{x^2+1} + \frac{B}{x-1}$

b. $\frac{A}{x^3} + \frac{B}{x}$

c. $\frac{Ax+B}{x^2} + \frac{C}{x-1}$

d. $\frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$ ✓

$x^3 - x = x(x-1)(x+1)$: three linear factors.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Sum of series $\sum_{n=2}^{\infty} \frac{1}{n^2-1}$ (partial fractions $\frac{1}{2}(\frac{1}{n-1} - \frac{1}{n+1})$):

a. 1

b. 3/4 ✓

c. 3/2

d. 5/4

Telescoping sum $= \frac{1}{2}(1 + 1/2) = 3/4$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Differential Equations

Partial fractions for $\frac{1}{(x+1)^2}$ uses:

a. **repeated linear factor** ✓

b. linear factors

c. irreducible quadratic

d. none

$\frac{A}{x+1} + \frac{B}{(x+1)^2}$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 12.01 Poisson Distribution

Consider the region R in the xy-plane bounded by the curves $y = x^2$ and $y = 4$. What is the area of this region?

a. $\frac{32}{3}$ ✓

b. $\frac{16}{3}$

c. 8

d. 16

The area of the region bounded by two curves $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$, where $f(x) \geq g(x)$ on $[a, b]$, is given by the integral $\int_a^b (f(x) - g(x))dx$. First, we find the intersection points of $y = x^2$ and $y = 4$ by setting them equal: $x^2 = 4$, which gives $x = -2$ and $x = 2$. So, our limits of integration are $a = -2$ and $b = 2$. On the interval $[-2, 2]$, the line $y = 4$ is above the parabola $y = x^2$. Therefore, the area A is given by: $A = \int_{-2}^{2} (4 - x^2)dx$ Now, we eva.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 13.01 Continuous Random Variables

Consider the quadratic function $f(x) = x^2 - 8x + k$. The graph of $y = f(x)$ is a parabola that intersects the x-axis at the points $(2, 0)$ and $(6, 0)$. Find the value of k.

a. **12** ✓

b. 16

c. 20

d. 24

Since the parabola intersects the x-axis at $(2, 0)$ and $(6, 0)$, these are the roots of the quadratic equation $f(x) = 0$. Therefore, we can write the function in factored form as $f(x) = a(x - 2)(x - 6)$. Expanding this, we get $f(x) = a(x^2 - 8x + 12)$. Comparing this with the given function $f(x) = x^2 - 8x + k$, we can see that the coefficient of x^2 is 1. Thus, $a = 1$. So, $f(x) = x^2 - 8x + 12$. By comparing the constant terms, we find that $k = 12$. Alternatively, we can substitute one of the roots into the o.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 14.01 Sampling & Estimation

Consider the slope field shown below. Which of the following differential equations could represent this slope field, assuming the slopes are defined at all points except where division by zero occurs? $\frac{dy}{dx} = \frac{x}{y}$

a. $\frac{dy}{dx} = \frac{x}{y}$ ✓

b. $\frac{dy}{dx} = \frac{y}{x}$

c. $\frac{dy}{dx} = x - y$

d. $\frac{dy}{dx} = x + y$

The slope field shows that at points where $y = 0$ (the x-axis), the slopes are undefined (vertical lines). This is consistent with the differential equation $\frac{dy}{dx} = \frac{x}{y}$, as division by zero occurs when $y = 0$. Additionally, observe the slopes in different quadrants. In the first quadrant ($x > 0, y > 0$), the slopes are positive, consistent with $\frac{x}{y} > 0$. In the second quadrant ($x0$), the slopes are negative, consistent with $\frac{x}{y}0$. In the fourth quadrant ($x > 0, y < 0$), the slopes are.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 15.01 Hypothesis Testing

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. Determine the sum of the x-intercepts.

a. **4** ✓

b. 3

c. 1

d. -3

The x-intercepts of a function are the values of x for which $f(x) = 0$. For the quadratic function $f(x) = x^2 - 4x + 3$, we can find the x-intercepts by solving the equation $x^2 - 4x + 3 = 0$. This is a quadratic equation that can be factored. We are looking for two numbers that multiply to 3 and add to -4. These numbers are -1 and -3. Therefore, the equation can be factored as $(x - 1)(x - 3) = 0$. The solutions are $x = 1$ and $x = 3$. These are the x-intercepts. The sum of the x-intercepts is $1 + 3 = 4$. Alternative.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Exponentials & Logarithms

The graph of the function $g(x) = x^2 - 6x + k$ is shown below. The graph intersects the x-axis at points P and Q. Point P has coordinates $(2, 0)$. Find the value of the parameter k.

a. 1

b. **2** ✓

c. 4

d. 8

The graph of the quadratic function $g(x) = x^2 - 6x + k$ intersects the x-axis at points where $g(x) = 0$. We are given that one of these intersection points is P, with coordinates $(2, 0)$. This means that when $x = 2$, $g(x) = 0$. We can substitute these values into the function to find the value of k: $g(2) = (2)^2 - 6(2) + k$ $0 = 4 - 12 + k$ $0 = -8 + k$ $k = 8$ Therefore, the value of the parameter k is 8.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Advanced Trigonometry

Consider the quadratic function $f(x) = x^2 - 4x + 3$. Find the value of x where the graph of $f(x)$ intersects the x-axis at the point $(x, 0)$, given that $x > 1$.

a. 2

b. **3** ✓

c. 4

d. 5

The graph of $f(x)$ intersects the x-axis when $f(x) = 0$. We set the function equal to zero and solve for x: $x^2 - 4x + 3 = 0$ Factoring the quadratic equation, we get: $(x - 1)(x - 3) = 0$ This gives two possible solutions for x: $x = 1$ or $x = 3$. The question specifies that we need to find the value of x where $x > 1$. Therefore, the correct value is $x = 3$.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Differentiation

Consider the solid E bounded by the surfaces $z = x^2 + y^2$ and $z = 8 - x^2 - y^2$. Calculate the volume of this solid using a triple integral by first transforming to cylindrical coordinates. Express your answer in terms of π.

a. $\frac{16\pi}{3}$

b. $\frac{32\pi}{3}$ ✓

c. 16π

d. $\frac{64\pi}{3}$

The solid E is bounded below by the paraboloid $z = x^2 + y^2$ and above by the paraboloid $z = 8 - x^2 - y^2$. To find the intersection of these two surfaces, we set them equal: $x^2 + y^2 = 8 - x^2 - y^2$, which simplifies to $2(x^2 + y^2) = 8$, or $x^2 + y^2 = 4$. This is a circle of radius 2 in the xy-plane. In cylindrical coordinates, $x = r\cos(\theta)$, $y = r\sin(\theta)$, and $z = z$. The equations of the surfaces become $z = r^2$ and $z = 8 - r^2$. The intersection occurs at $r^2 = 8 - r^2$, so.

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